

Integrated Reservation System

Project Requirements & Solution Design Document

Prepared For	Travel / Booking Platform Team
Product Type	B2C + B2B Reservation Marketplace
Core Domains	Flights, Bus, Hotel, Hostel, Tours, Referrals, Coupons, Orders
Document Version	v1.0
Purpose	Use as project scope, execution brief, and development reference

This document defines the business scope, features, workflows, architecture, data model, roles, integrations, and delivery priorities for a full-scale reservation ecosystem inspired by large travel and ticketing marketplaces.

1. Executive Summary

The proposed platform is an integrated reservation system where customers can search, compare, and reserve flights, bus tickets, hotels, hostels, and tour packages from a single application. The solution also includes coupon management, referral rewards, order lifecycle management, payment handling, cancellation and refund processing, supplier operations, and admin monitoring.

The target product is not a simple booking website; it is a marketplace-grade travel commerce platform that unifies inventory aggregation, booking management, promotional growth, payment reconciliation, customer support, and analytics in one ecosystem.

2. Business Goals

- Enable customers to reserve multiple travel products from one platform with a consistent checkout experience.
- Support both direct inventory and third-party supplier integrations for scalable growth.
- Increase conversion through coupons, referrals, personalized offers, and cross-sell bundles.
- Provide strong post-order support including modification, cancellation, refunds, disputes, and wallet adjustments.
- Give operations and admin teams complete visibility into orders, suppliers, commissions, fraud risks, and performance.

3. Product Scope

Domain	Included Features	Key Business Value
Flights	Search, fare details, traveler info, seat/baggage add-ons, booking, cancellation, refund	High-demand travel inventory with dynamic pricing
Bus	Route search, boarding/drop points, seat selection, schedule comparison, ticket issuance	Mass-market intercity reservation use case
Hotel / Hostel	Room search, occupancy rules, property details, booking, partial refunds, reviews	Accommodation monetization with upsell potential
Tours	Tour listing, itinerary details, date slots, traveler categories, booking, supplier confirmation	Package commerce and bundled travel sales
Promotions	Coupons, referral rewards, wallet credits, campaign targeting	Growth, retention, and repeat orders
Commerce	Cart, checkout, payments, order splits, invoices,	Unified transactional backbone

	cancellations, refunds	
Operations	Supplier panel, admin control, reconciliation, support tools, reports	Scalable business operations

4. User Roles

Role	Description	Main Permissions
Customer	End user who searches and places reservations	Search, reserve, pay, cancel, review, use coupon, refer friends
Supplier / Vendor	Inventory provider or partner operator	Manage listings, rates, availability, booking confirmations, support requests
Agent / B2B Reseller	Optional partner booking on behalf of travelers	Create customer bookings, manage markups, track commissions
Support Staff	Customer support and operations team	View orders, process changes, trigger refund workflows, raise cases
Admin	System owner and governance layer	User management, promotion control, reconciliation, analytics, fraud review
Finance Admin	Settlement and money movement control	Payment review, supplier payout, coupon liability, refund audit

5. Functional Modules

5.1 Authentication & Account Management

- Email / phone registration, OTP verification, password login, social login (optional).
- Profile management with traveler details, saved passengers, saved addresses, and emergency contact info.
- Role-based access control for customer, supplier, agent, support, and admin users.
- Session security with JWT access token, refresh token, device log tracking, and suspicious login detection.

5.2 Search & Discovery Engine

- Unified search entry with category-based flows for flights, bus, hotels, hostels, and tours.
- Filters for price, rating, amenities, departure timing, airline/bus operator, refundable type, stay type, and package duration.
- Sorting by lowest price, best value, shortest duration, highest rating, recommended, and most popular.

- Search result caching and analytics capture for demand trends and abandoned intent analysis.

5.3 Flight Reservation Module

- One-way, round-trip, and multi-city support.
- Fare class visibility, baggage allowance, layover details, and traveler policy notes.
- Passenger information capture, contact details, ancillary add-ons, booking confirmation, and e-ticket issuance.
- Cancellation policy, reschedule request flow, and refund status tracking.

5.4 Bus Reservation Module

- Route and schedule search by source, destination, and date.
- Seat map selection, operator details, boarding/drop point selection, and traveler assignment.
- Ticket issuance with QR/PDF support and trip status tracking.
- Operator cancellation handling, replacement options, and refund initiation.

5.5 Hotel & Hostel Reservation Module

- Property listing pages with images, amenities, house rules, check-in/check-out policy, map location, and review scores.
- Inventory by room type, occupancy, breakfast plan, cancellation type, and nightly pricing.
- Property reservation, guest details, special requests, and booking voucher generation.
- Hostel-specific bed inventory, dorm rules, and shared facility information.

5.6 Tour Package Module

- Package listing with itinerary, inclusions/exclusions, images, departure dates, duration, pickup points, and guide notes.
- Traveler category pricing such as adult, child, infant, student, or private group.
- Booking request, supplier confirmation, slot management, and voucher issuance.
- Optional add-ons such as transport, meals, tickets, and insurance.

5.7 Referral System

- Each eligible user receives a referral code or share link.
- Reward rules for referrer and referee, including conditions such as first completed order only.
- Wallet credit or coupon issuance after order validation.
- Fraud prevention controls for self-referral, duplicate device abuse, and fake account loops.

5.8 Coupon & Promotion System

- Global coupons, category-specific coupons, supplier-funded offers, and user-segment campaigns.
- Rules by minimum amount, date range, first order, payment method, destination, product type, and usage limit.
- Stacking rules, conflict resolution, and budget cap monitoring.

- Real-time coupon validation at cart and checkout stages.

5.9 Cart, Checkout & Order System

- Single-product and multi-product cart support.
- Order split handling when multiple suppliers or products exist in one checkout.
- Price breakdown including base fare, taxes, service fees, discounts, insurance, and convenience charges.
- Order creation, payment capture, confirmation, invoice generation, and status history.

5.10 Payment, Wallet & Refund Management

- Payment gateway integration, pending transaction reconciliation, and failure retry handling.
- Partial or full refund processing depending on supplier rules and platform policy.
- Internal wallet / credit ledger for referral rewards, compensation, and refund adjustments.
- Refund states such as requested, approved, supplier_pending, processed, failed, and reversed.

5.11 Review & Rating System

- Only eligible travelers with completed bookings can leave reviews.
- Star ratings, media upload, moderated comments, and property/service quality tagging.
- Review abuse checks and admin moderation queue.
- Review aggregation on listing pages and supplier dashboards.

5.12 Notifications & Communication

- Transactional email, SMS, and in-app notifications for booking, payment, cancellation, refund, and reminders.
- Pre-departure travel reminders and post-stay feedback requests.
- Admin-configurable templates with placeholder variables and event triggers.

5.13 Supplier / Vendor Panel

- Inventory creation and updates, availability calendars, pricing rules, black-out dates, and booking queue.
- Booking acceptance where required, cancellation handling, support messaging, and settlement summary.
- Performance views such as revenue, occupancy, conversion, and cancellation rate.

5.14 Admin & Operations Console

- Global dashboard, user management, supplier approval, offer management, case management, and analytics.
- Manual intervention tools for order correction, goodwill coupon issue, and fraud flagging.
- Audit logs for administrative actions and configurable access permissions.

6. Core Booking Workflow

- User selects product category and enters search criteria.
- System fetches inventory from internal database and/or external supplier APIs.

- User applies filters, views details, selects product, and enters traveler/guest data.
- Coupon and wallet validation is performed before final amount is shown.
- Payment is authorized or captured through gateway.
- Order is created with one master order and one or more child bookings as needed.
- Supplier booking confirmation is received or queued for asynchronous confirmation.
- Voucher / ticket / confirmation email is generated and delivered to user.
- Order timeline remains available for support, cancellation, modification, and refund actions.

7. Order Lifecycle Design

Recommended master order statuses: draft, payment_pending, paid, booking_in_progress, confirmed, partially_confirmed, failed, cancelled, refund_pending, refunded, completed, disputed.

Recommended child booking statuses: pending_supplier, confirmed, ticketed, voucher_issued, partially_cancelled, cancelled, reschedule_requested, refund_in_process, refund_completed.

All status changes must be event-driven and logged with timestamp, actor, previous status, new status, and reason note.

8. Key Real-World Problem Statements to Solve

- Double booking risk when inventory changes between search and payment.
- Payment success but supplier booking failure due to stale fare or sold-out inventory.
- Different cancellation rules per supplier, product type, and timing window.
- Referral abuse, coupon overuse, fake accounts, and promotional liability leakage.
- Multi-supplier checkout where one booking succeeds and another fails.
- Support difficulty if order history, payment logs, and supplier communication are fragmented.
- Settlement disputes between platform and suppliers due to commission, discounts, or refund adjustments.
- Need for complete audit trail for admin actions, financial events, and booking changes.

9. Non-Functional Requirements

Area	Requirement	Notes
Performance	Search APIs should respond quickly under high concurrency	Use caching, pagination, and query optimization
Security	Protect auth, payments, PII, and admin actions	JWT, RBAC, rate limiting, encryption, audit logs
Scalability	Support category growth and partner expansion	Modular services and supplier abstraction layer
Reliability	Prevent order inconsistency during failures	Idempotency keys, retries, queue workers, reconciliation jobs

Observability	Track system health and business events	Centralized logs, metrics, alerts, tracing
Usability	Booking flow should remain simple despite platform complexity	Clear UI states, validation, and recovery messaging

10. Suggested Technical Architecture (MERN)

Frontend: React.js or Next.js web application with modular booking flows, reusable search components, role-based dashboards, and state management for cart, pricing, and session flows.

Backend: Node.js + Express.js REST APIs (or modular API gateway pattern), authentication service, booking service, inventory service, promotion service, payment service, notification service, reporting layer, and background workers.

Database: MongoDB for transactional and catalog data, with deliberate schema design for orders, users, coupons, referrals, inventory snapshots, supplier mappings, and audit events.

Supporting Infrastructure: Redis for caching and temporary holds, message queue for asynchronous confirmation and notifications, object storage for media, and third-party gateway/API integrations.

11. Recommended Collections / Entities

- users
- roles
- permissions
- travelers
- suppliers
- supplier_accounts
- flight_routes / providers
- bus_routes / operators
- properties
- rooms / beds
- tour_packages
- inventory_snapshots
- search_logs
- carts
- orders
- order_items
- payments
- refunds
- wallet_transactions
- coupons
- coupon_redemptions

- referrals
- reviews
- notifications
- support_tickets
- audit_logs
- settlements
- commissions

12. Integration Requirements

- Payment gateway integration for card, bank, wallet, or local payment methods.
- Supplier APIs for flight/bus/hotel/tour inventory where dynamic booking is required.
- Email, SMS, and push notification providers.
- Cloud storage for property images, vouchers, and support attachments.
- Analytics and logging tools for product and operational monitoring.

13. Admin Reporting & Analytics

- GMV, confirmed revenue, supplier-wise bookings, top routes, top destinations, occupancy trends, cancellation ratio, refund volume, coupon burn, referral conversion, repeat user rate.
- Funnel analytics: search -> detail view -> cart -> checkout -> payment -> confirmation.
- Operational reports: pending supplier confirmations, failed payments, disputed refunds, over-budget campaigns, dormant suppliers.
- Finance reports: payout due, commission earned, refunded amount, wallet liabilities, promotional liabilities.

14. Security & Compliance Controls

- Password hashing, token rotation, secure cookie strategy where applicable, and brute-force protection.
- Input validation and sanitization on all request bodies, query params, and admin tools.
- Role-based permissions at API and UI levels.
- PII masking in logs and restricted access to financial actions.
- Audit log for admin changes, refund approvals, supplier settlement updates, and promotion edits.

15. Recommended Delivery Phases

Phase	Goal	Main Deliverables	Priority
Phase 1	Core Foundation	Auth, roles, search shell, cart, checkout, orders, payment integration basics	Critical

Phase 2	Booking Domains	Flights, bus, hotel/hostel, tours, supplier panel basics	Critical
Phase 3	Growth Features	Coupons, referrals, wallet credits, review system, notifications	High
Phase 4	Operations & Control	Admin console, reconciliation, support tools, reports, fraud checks	High
Phase 5	Optimization	Scalability improvements, analytics depth, personalization, recommendation engine	Medium

16. Suggested Evaluation Criteria for the Developer

- Clean architecture and modular folder structure.
- Robust API design with validation, error handling, and consistent response patterns.
- Thoughtful MongoDB schema design and aggregation usage for reporting.
- Real handling of booking edge cases, refund states, and promotion rules.
- Security, observability, and audit readiness.
- Production-grade documentation including ERD, API collection, setup guide, and deployment notes.

17. Final Recommendation

This project should be treated as a marketplace-grade travel commerce system rather than a basic CRUD application. The developer should be asked to build both customer-facing flows and internal operational controls. A strong implementation will demonstrate not only MERN stack proficiency but also product thinking, system design understanding, business rule handling, and real-world edge-case management.